

Department of Statistics

University of Lagos

STA 111 (Descriptive Statistics)
Lecture Note

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Topics

- ☐ Skewness
- ☐ Kurtosis

STA111 LECTURE SLIDE

Skewness

- A distribution is said to be symmetry if it is possible to cut its graph into two mirror image halves.
- Such distributions have bell-shape graphs. This shape of the frequency curves are characterized by the fact that observations equidistant from the central maximum have the same frequency e.g the normal curve.
- Skewness is the degree of asymmetry (departure from symmetry) of a distribution. If the frequency curve (smoothed frequency polygon) of a distribution has the length of one of its tails (relative to the central section).
- The concept of skewness would be clear from the following three diagrams showing a symmetrical distribution, a positively skewed distribution and negatively skewed distribution.

A Symmetrical Distribution

(i) Symmetrical Distribution

It is clear from the diagram below that in a symmetrical distribution the values of mean median and mode coincide. The spread of the frequencies is the same on both sides of the center point of the curve.

(ii) Asymmetrical Distribution

A distribution which is not symmetrical is called a skewed distribution and such a distribution could either be positively skewed or negatively skewed as would be clear from the following diagram

(iii) Positively Skewed Distribution

In the positively skewed distribution, the value of mean is maximum and that of the mode is minimum – the median lies in between the two.

A Symmetrical Distribution (Cont'd.)

(iv) Negatively Skewed Distribution

In a negatively skewed distribution, the value of the mode is maximum and that of mean minimum – the median lies in between the two.

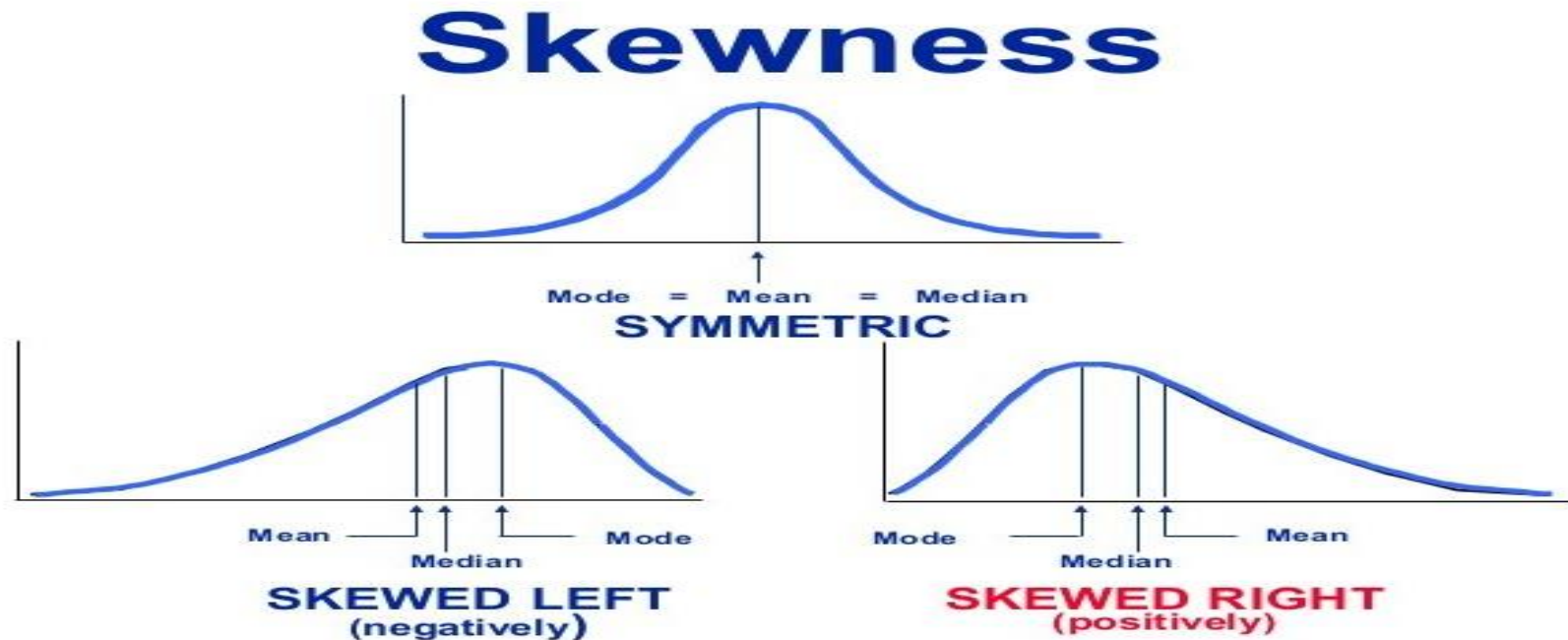


Figure 1a. Symmetric, Negative Skewed and Positive Skewed

Skewness Diagram

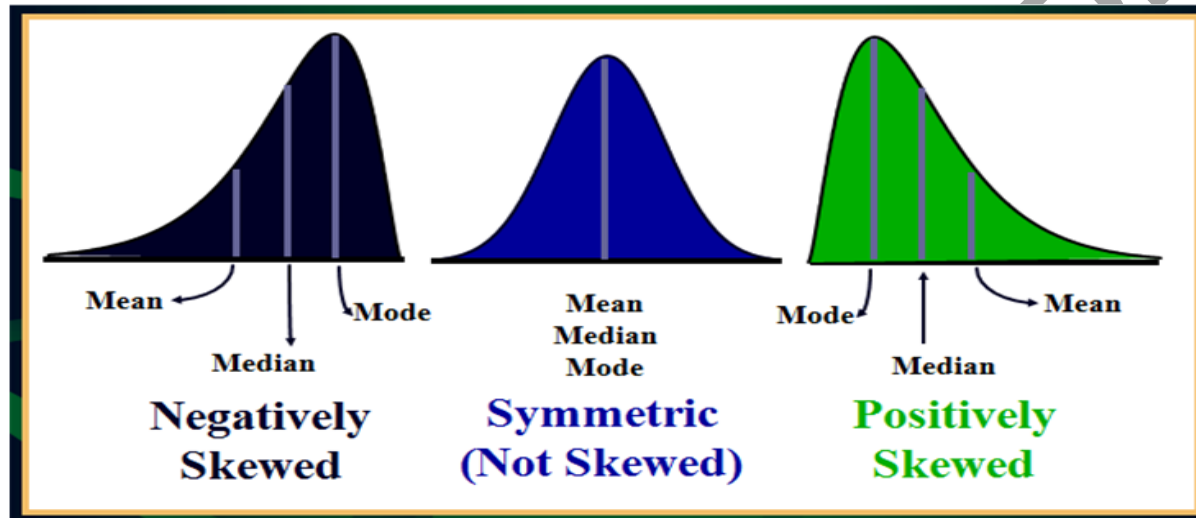


Figure 1b. Plot of Symmetry, Positive and Negative Skewed

Test of Skewness

In order to ascertain whether a distribution is skewed or not, the following tests may be applied. Skewness is present if:

- The values of mean, median, and mode do not coincide.
- When the data are plotted on a graph they do not give the normal bell-shaped form i.e when cut along a vertical line through the centre the two halves are not equal.

Skewness (Cont'd.)

Frequencies are not equally distributed at points of equal deviation from the mode. Conversely stated, when skewness is absent, i.e. in case of a symmetrical distribution, the following conditions are satisfied:

- The values of mean, median, and mode coincide.
- Data when plotted on a graph give the normal bell-shaped form.
- Frequencies are equally distributed at points of equal deviation from the mode.

Skewness (Cont'd)

- **Skewness (SK)** = $\frac{\sum (x - \bar{x})^3}{(n-1) \cdot s^3}$ **or** $\left(\frac{\sum (x - \bar{x})^3}{n-1} \right) / s^3$

where $\bar{x} = \frac{\sum x}{n}$

However, for ungrouped data

- **Skewness (SK)** = $\frac{\sum f(x - \bar{x})^3}{(\sum f - 1) \cdot s^3}$ **or** $\left(\frac{\sum f(x - \bar{x})^3}{\sum f - 1} \right) / s^3$

where $\bar{x} = \frac{\sum f(x)}{\sum f}$,

Kurtosis

Kurtosis is a statistical number that tells us if a distribution is taller or shorter than a normal distribution. If a distribution is similar to the normal distribution, the Kurtosis value is 0. If Kurtosis is greater than 0, then it has a higher peak compared to the normal distribution. If Kurtosis is less than 0, then it is flatter than a normal distribution.

There are three types of distributions:

- **Leptokurtic:** Sharply peaked with fat tails and less variable.
- **Mesokurtic:** Medium peaked
- **Platykurtic:** Flattest peak and highly dispersed.

Example of different types of kurtosis

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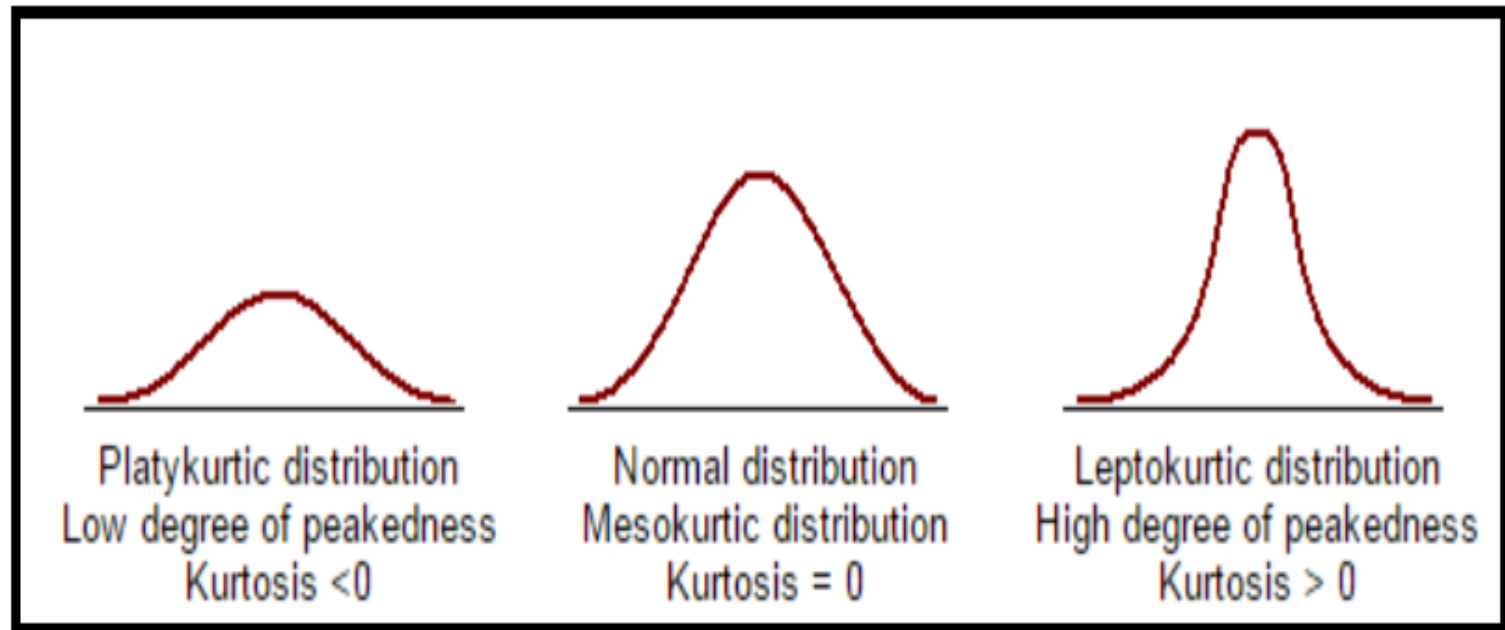


Figure 1c. Plot of different types of Kurtosis

Kurtosis Cont'd.

The Formula of kurtosis is:

- **Kurtosis (K)** = $\frac{\sum(x-\bar{x})^4}{(n-1) \cdot S^4}$ **or** $\left(\frac{\sum(x-\bar{x})^4}{n-1}\right) / S^4$

where $\bar{x} = \frac{\sum x_i}{n}$

For ungroup data

- **Kurtosis (K)** = $\frac{\sum f(x-\bar{x})^4}{(\sum f-1) \cdot S^4}$ **or** $\left(\frac{\sum f(x-\bar{x})^4}{\sum f-1}\right) / S^4$

where $\bar{x} = \frac{\sum f(x)}{\sum f}$,

For grouped data

Example

Suppose we have the following ungroup data: 7, 9, 10, 8, 6

- Solution

x	$(x - \bar{x})$	$(x - \bar{x})^2$	$(x - \bar{x})^3$	$(x - \bar{x})^4$
7	-1	1	-1	1
9	1	1	1	1
10	2	4	8	16
8	0	0	0	0
6	-2	4	-8	16
Total		10	0	34

$$\bar{x} = \frac{\sum x_i}{n} = \frac{7 + 9 + 10 + 8 + 6}{5} = 8$$

Example Cont'd.

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1} = \frac{10}{4} = 2.5$$

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}} = \sqrt{2.5} = 1.58$$

$$\text{Skewness (SK)} = \frac{\sum(x - \bar{x})^3}{(n - 1) \cdot s^3} = \frac{0}{4 \times 1.58^3} = 0$$

$$\text{or } \left(\frac{\sum(x - \bar{x})^3}{n - 1} \right) / s^3 = \left(\frac{0}{4} \right) / 1.58^3 = 0$$

Hence, **Skewness = 0** and we can conclude that the distribution is **Symmetrical**.

Example Cont'd.

For Kurtosis, we have

$$\text{Kurtosis } (K) = \frac{\sum(x - \bar{x})^4}{(n - 1) \cdot S^4} = \frac{34.0}{4 \times 1.58^4} = \frac{34.0}{15.78} = 1.36$$

$$\text{or } \left(\frac{\sum(x - \bar{x})^4}{n - 1} \right) / S^4 = \left(\frac{34.0}{4} \right) / 1.58^4 = 1.36$$

Then, Kurtosis = 1.36 and we can conclude that the distribution is Leptokurtic distribution.

Example of Grouped Data

- Calculate sample skewness and kurtosis from the following grouped data

Class Interval	2 – 4	4 – 6	6 – 8	8 – 10
Frequency	3	4	2	1

Example Cont'd.

Solution

<i>Class Int</i>	<i>f</i>	<i>Mid point (x)</i>	<i>f(x)</i>	<i>(x - $\bar{x}$)</i>	<i>f(x - $\bar{x}$)²</i>	<i>f(x - $\bar{x}$)³</i>	<i>f(x - $\bar{x}$)⁴</i>
2 - 4	3	3	9	-2.2	14.52	-31.944	70.27
4 - 6	4	5	20	-0.2	0.16	-0.032	0.0064
6 - 8	2	7	14	1.8	6.48	11.664	20.98
8 - 10	1	9	9	3.8	14.44	54.872	208.50
Total	10	--	52		35.6	34.56	299.79

$$\text{Mean} = \bar{x} = \frac{\sum f(x)}{\sum f} = \frac{52}{10} = 5.2$$

$$s^2 = \frac{\sum f(x - \bar{x})^2}{\sum f} = \frac{35.6}{10} = 3.56$$

$$s = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} = \sqrt{3.56} = 1.89$$

Example Cont'd.

$$\text{Skewness (SK)} = \frac{\sum f(x - \bar{x})^3}{(\sum f - 1) \cdot S^3} = \frac{34.56}{9 * 1.89^3} = 0.57$$

Hence, skewness = 0.57. It is a right skewed distribution.

$$\text{Kurtosis (K)} = \frac{\sum f(x - \bar{x})^4}{(\sum f - 1) * S^4} = \frac{299.79}{9 * 1.89^4} = 2.61$$

Kurtosis = 2.61, this implies that the distribution of the data is a leptokurtic distribution.

End of Lecture.
Any Question so far?